

C.3 Comparison of Radiologists to Original Reads

The reports from the radiologists who originally read the patient cases included in our sample were classified as positive/negative/uncertain for each pathology using AI predictions generated by the CheXbert algorithm described in [Smit et al. \(2020\)](#). We compare the accuracy of the original reads relative to the ground truth with the radiologists in our sample under the treatment arm with clinical history and no AI assistance. We do this for each pathology by converting the probability reports elicited during the experiment to positive/negative assessments, where positive is defined as having a probability greater than 50%. We convert the CheXbert labels to positive/negative assessments by including the uncertain cases as positive.³³ We then calculate the accuracy of the experiment reads and the CheXbert labels for groups of pathologies focused on in this study and test the null hypothesis that the accuracy of the radiologists is the same. The results of this analysis are in Table C.5, and those for when treating uncertain cases negative are in Table C.6.

Table C.5: Comparing Experiment Assessments to Original Reads

	Top-Level with AI (1)	Pooled with AI (2)	Abnormal (3)
EXPERIMENT	-0.004 (0.016)	-0.005 (0.006)	-0.086 (0.025)
CONSTANT	0.194 (0.016)	0.090 (0.006)	0.466 (0.028)
OBSERVATIONS	9718	53449	4859
R-SQUARED	0.000	0.000	0.002

Note: Regression of indicator equal to one if binarized assessment is equal to ground truth from both the original reads and experiment reads onto a constant and an indicator equal to one if the radiologist was in the experiment. Standard errors are clustered at the patient-case level.

³³For all pathologies but bacterial pneumonia and atelectasis, fewer than 5% of patients have uncertain cases. For abnormal and all of the top-level pathologies with AI, there are no cases with uncertain labels.

Table C.6: Comparing Experiment to Original Reads: Uncertain as Not Present

	Top-Level with AI (1)	Pooled with AI (2)	Abnormal (3)
EXPERIMENT	-0.004 (0.016)	0.020 (0.005)	-0.086 (0.025)
CONSTANT	0.194 (0.016)	0.065 (0.005)	0.466 (0.028)
OBSERVATIONS	9718	53449	4859
R-SQUARED	0.000	0.000	0.002

Note: Regression of indicator equal to one if binarized assessment is equal to ground truth from both the original reads and experiment reads onto a constant and an indicator equal to one if the radiologist was in the experiment. Standard errors are clustered at the patient-case level.

C.4 Robustness

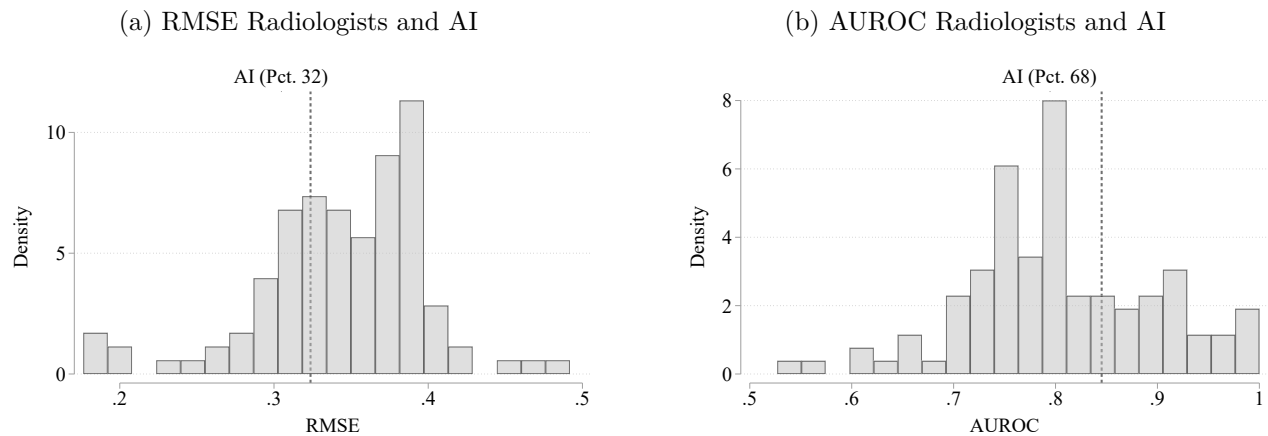
In this section, we show the robustness of the results from Section 4.2.

C.4.1 By Design

Design 1

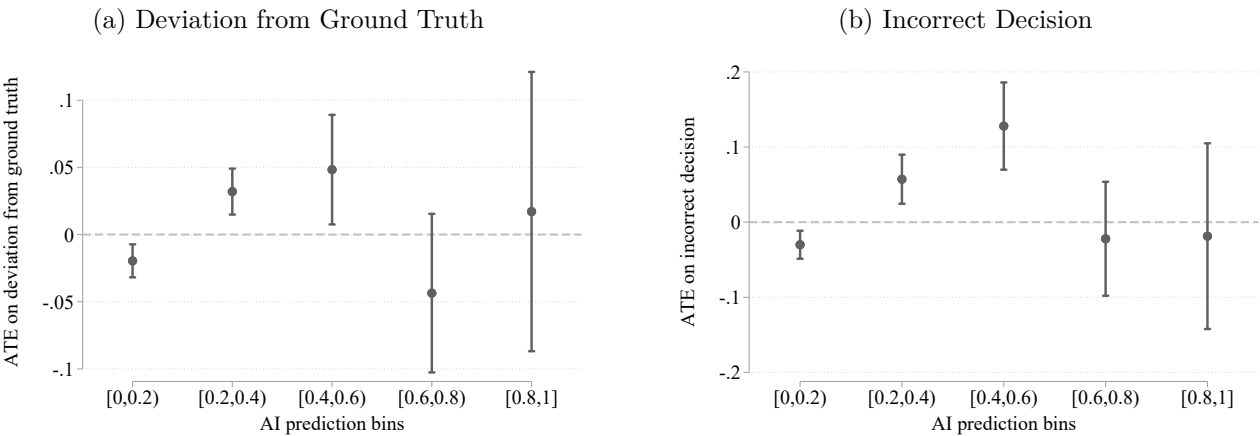
This section presents summaries of radiologist accuracy and treatment effect estimates using data from design 1. We do not present treatment effects conditional on radiologist prediction as same cases are not read in the design.

Figure C.12: Comparing AI performance to Radiologists



Note: Main specifications similar to figure 1 with the exception that observations are from design 1 only.

Figure C.13: Conditional treatment effect given AI prediction



Note: Main specifications similar to figure 3 with the exception that observations are from design 1 only.